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**Collision Severity Analysis In Canadian Road Accidents**

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## Abstract

Road traffic accidents are a major public safety issue in Canada. Road traffic accidents cause substantial social, economic, and health impacts each year. Despite improvements in vehicle safety and traffic regulations, motor vehicle collisions continue to result in significant number of deaths and injuries annually. According to recent national statistics report, more than 1,900 fatalities and more than 118,000 injuries in 2022. highlighting the ongoing need for improved road safety strategies.

This study analyzes the factors that influence the severity of injuries in Canadian road accidents using the publicly available National Collision Database (NCDB) provided by Transport Canada. The dataset containing over 5.8 million accident records with 22 variables, includes information on environmental conditions, roadway characteristics, driver demographics, and vehicle-related attributes, with observations covering the period 1999 to 2014.

To identify the key determinants of injury severity, this research applies binary logistic regression and ordinal logistic regression models estimated using Maximum Likelihood Estimation (MLE). Model performance is evaluated using several metrics including accuracy, balanced accuracy, ROC–AUC, and F1 score.

The results indicate that environmental conditions, road infrastructure characteristics, and human factors play an important role in determining injury severity in road traffic accidents. In particular, hazardous road surfaces, adverse weather conditions, and certain road-user categories are associated with higher probabilities of severe injuries. Scenario-based simulations are conducted to examine how injury probabilities change under different crash situations. In addition, Monte Carlo simulations are used to study the uncertainty and variation in crash risk across thousands of possible scenarios.

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The results of this study help in understanding the key factors that affect accident severity in Canadian road accidents.

The findings can offer practical information to policy makers and can be used in the formulation of evidence-based policies to enhance road safety outcomes in Canada.

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## Chapter 1: Introduction

### 1.1 Background

Road traffic accidents continue to be one of the significant public safety challenges worldwide. They cause loss of life, physical injuries, emotional trauma, damage to property, vast economic costs, burdens on individuals and families, health-care systems and governments. In Canada, road collisions still, remain a significant social and economic burden with all the efforts made to improve transportation systems infrastructure, traffic control mechanisms and vehicle safety technologies. While many crashes are relatively minor in severity, some result in serious or fatal injuries; crash severity is therefore an important issue for transportation safety research and public policy.

The knowledge of how severity of injuries during road accidents is determined is necessary in the development of effective safety measures. The results of crashes can hardly be attributed to one factor. As an alternative, they are determined by a mixture of environmental factors, road features, traffic control scenario, human factor, and vehicle factors. As an illustration, poor weather, unsafe road surface, poor road alignment, absence of quality road managing measures, susceptible road-user position, and inadequate safety apparatus can all serve to escalate worse results of injury. The complexity of the situation means that the severity of crash must be examined through combined data-driven methodology instead of depending on descriptive statistics only.

The increasing availability of large-scale datasets of traffic collisions has opened up new potentials to study crash severity in terms of statistical modeling and simulation methodologies. Such approaches can go beyond the number of crashes and offer more in-depth information

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about the influence of certain variables on the likelihood of getting injured. Specifically, predictive models may be used to learn the direction and strength of the relationships between risk factors and injury outcomes, and scenario analysis and simulation may be used to understand the dynamics of crash risk variations under varying conditions. This kind of evidence can be helpful in road safety planning, infrastructure enhancement and policy formulation.

This capstone project is dedicated to collision and severe injuries analysis in a Canadian road accident with the help of advanced data analytics tools. The data used in the study is the large accident dataset consisting of millions of observations and variables being weather, road surface, road alignment, traffic control, road-user type, safety equipment, age, sex, vehicle type, and vehicle age. Binary logistic regression and ordinal logistic regression are used to model the severity of injury and further simulation-based techniques are used to analyze how the risk is predicted to vary under a chosen crash condition. This study will offer predictive and practical suggestions to enhance road safety in Canada by merging statistical analysis and scenario simulation, as well as Monte Carlo techniques.

### 1.2 Problem Statement

The number of injuries and fatalities due to road accidents in Canada still remains high, posing significant social, economic, and public health problems. Despite the fact that government reports may include descriptive summaries of crash trends, they tend to address counts and percentages of crashes and pay little attention to the numerous risk factors which contribute to injury severity. This leaves still to be done some deeper analytical research that can determine which factors are most highly correlated with severe crash outcomes and how these factors combine to modulate the risk.

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One of the biggest problems of road safety analysis is that multiple domains simultaneously influence the severity of a crash. The weather and road surface are some of the environmental conditions that might lead to an increase in the risk of crashing, however, human and vehicle factors can also play a significant part in the ultimate outcome. As an example, a crash of a vulnerable road user under hazardous road conditions may have a greatly increased probability of injury as compared to a crash under normal conditions. As such, variables cannot be analyzed independently. A better solution is to create statistical models, which can assess the joint effects of environmental, traffic, human, and vehicle-related factors.

The second issue is that policymakers require actionable insights rather than only descriptive summaries. It is helpful to know that crashes occur more frequently under some circumstances, but the more important point is to estimate the extent to which risk of injury grows when road surface is unsafe, when there is no signal control traffic, or when crashes involve more vulnerable categories of road users. This kind of evidence can help infrastructure planning, safety regulation, and targeted intervention programs directly.

In these terms, the main issue that is to be resolved in this study is the necessity to find out and measure the major factors of the crash injury severity in Canadian road accidents through predictive modeling and analyzing it with the help of simulation. This investigation aims to find out what factors have the most significant effect on the risk of injuries and how these results can be converted into practical recommendations in order to enhance road safety.

### 1.3 Research Objectives

The primary aim of the research will be to evaluate the variables that determine the severity of injuries in Canadian road accidents through statistical modeling and simulation.

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The objectives of the study are:

1. To test the effects of environmental and traffic conditions factors including weather, road surface, road alignment, road traffic control, month of year, weekday, hour of day, and number of vehicles on the severity of crash injuries.
2. To check the relationship between human and vehicle factors like safety equipment, type of road-user, sex or gender, vehicle type, vehicle age, and age of person and severity of injury.
3. To compare the predictive performance of separate and combined models in order to determine the most effective one in explaining the injury outcomes.
4. To compare the predictive performance of binary logistic regression and ordinal logistic regression to the crash severity problem.
5. The objective of the study is to estimate the probability of injury due to changes in selected crash conditions and risk profiles by using scenario simulation and Monte Carlo analysis.
6. To develop useful suggestions that can be used by government agencies and transportation planners to minimize serious injuries in road crashes.

### 1.4 Research Questions

This study is guided by the following research questions:

RQ1: How do environmental and traffic context factors affect crash injury severity in Canadian road accidents?

RQ2: How do human and vehicle-related factors affect crash injury severity in Canadian road accidents?

RQ3: Does a combined model including environmental, traffic, human, and vehicle variables perform better than separate models?

RQ4: Between binary logistic regression and ordinal logistic regression, which modeling approach provides better predictive performance for crash severity analysis?

RQ5: How does predicted injury probability change under different simulated crash scenarios and risk profiles?

### 1.5 Research Hypotheses

Based on prior transportation safety literature and the logic of the study, the following hypotheses are proposed:

- Null Hypotheses ( $H_{01}$ ) : Environmental conditions, road context characteristics, and traffic control variables do not have a statistically significant effect on crash injury severity.
- Alternative Hypotheses ( $H_{11}$ ) : Environmental conditions, road context characteristics, and traffic control variables have a statistically significant effect on crash injury severity.
- Null Hypotheses ( $H_{02}$ ) : Human and vehicle-related factors do not significantly influence crash injury severity.
- Alternative Hypotheses ( $H_{12}$ ) : Human and vehicle-related factors significantly influence crash injury severity.
- Null Hypotheses ( $H_{03}$ ) : A combined model incorporating environmental, road context, human, and vehicle variables does not significantly improve the prediction of crash injury severity compared with separate models.
- Alternative Hypotheses ( $H_{13}$ ) : A combined model incorporating environmental, road context, human, and vehicle variables significantly improves the prediction of crash injury severity compared with separate models.
- Null Hypotheses ( $H_{04}$ ) : Simulation and Monte Carlo analysis do not provide meaningful insights into variations in crash injury probability across different risk scenarios.

- Alternative Hypotheses (H14) : Simulation and Monte Carlo analysis provide meaningful insights into variations in crash injury probability across different risk scenarios.

### 1.6 Significance of Study

This study has both theoretical and practical implications. In the academic context, it helps in studying crash severity by using predictive statistical modeling and simulation methods to determine injury outcomes of Canadian road accidents. Rather than using only descriptive statistics this study measures the magnitude of environmental, traffic, human and vehicle related factors on the likelihood of injury. It also compares binary and ordinal logistic regression methods to identify the most suitable model strategy to be used in the analysis of crash severity.

In practical terms, the results can be used to support evidence-based transportation safety policy. By highlighting some of the most important factors related to increased risk of injuries including dangerous road surfaces, unfavorable weather conditions, and traffic control scenarios, and the vulnerable groups of road users, the study offers information that can guide policymakers to prioritize safety efforts. These can involve a better maintenance of roads, better design of traffic control and better protection against pedestrians, cyclists and motorcyclists.

In addition, the analysis of a massive set of data and multiple analytical techniques, such as regression modeling, scenario simulation, sensitivity analysis, and Monte Carlo simulation, enable the study to present a more cohesive picture regarding the crash severity patterns and risk distribution.

### 1.7 Scope of Study

The data presented in this study concentrates on the Canadian road accident with variables of environmental factor, traffic, road-user factors, and vehicle factor. Only the variables represented in the chosen dataset are considered and analyzed according to the coding scheme



offered in that dataset. The severity of injury is modelled both in binary and ordinal framework, and further scenario-based simulation is applied to assess the impact of the chosen conditions on the probability of injury.

The research does not seek to determine causality, since the information used is not experimental but observational. Instead, the models provide statistical relationships between explanatory variables and injury outcomes. Further, though the dataset is large and comprehensive, there is a possibility that some potentially significant variables, including behavioral distraction, precise speed, geographic location description, or emergency response time, may not be fully captured. Despite these limitations, the study provides valuable insights into the determinants of crash injury severity and their potential implications for road safety policy.

### 1.8 Chapter summary

This chapter presented the background, problem statement, objectives, research questions, hypotheses, significance and scope of the study. It was determined that the severity of crash injuries is a multifactorial problem that may be affected by environmental, traffic, human, and vehicle causes, and predictive modelling can give useful results that are not always available through descriptive statistics. The second chapter summarizes the literature related and put this research into perspective with the broader research on the severity of traffic crashes.

## Chapter 2: Literature Review

### 2.1 Introduction

Road traffic accidents are a major public health concern across the world, and they result in millions of injuries, fatalities, and economic losses every year. Governments and transportation agencies continuously work to improve the safety of roads by making improvements in infrastructure, regulating policies and public awareness campaigns. However, understanding the factors that affect the severity of injuries in road crashes is a complex challenge because there are multiple interacting variables that affect crash outcomes, including environmental conditions, road infrastructure, human behavior, and vehicle characteristics.

Researchers have increasingly applied statistical and machine learning methods to understand crash severity and to identify the most important factors in determining injury outcomes. Among these methods, logistic regression models have been widely used since they allow researchers to estimate the probability of occurrences of injuries and evaluate the effect of different risk factors. In addition, simulation techniques such as scenario analysis and Monte Carlo simulation have been used to evaluate uncertainty and to help understand the variation of crash risks in different conditions.

This chapter reviews relevant literature on crash severity analysis, focusing on environmental and infrastructure factors, human and vehicle characteristics, statistical modeling techniques, and the use of simulation methods in road safety research.

### 2.2 Analysis of Crash Severity in Transportation Safety

Crash severity analysis is an attempt to understand the factors that affect the degree of injury during traffic accidents. Injury severity is usually no injury, injury, or fatal injury.

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Researchers have used different statistical methods to analyze crash data to determine what variables have the biggest impact on injury severity.

Savolainen et al. (2011) conducted a comprehensive review of statistical methods used in crash severity analysis and highlighted that crash outcomes are affected by a complex interaction of environmental, human, vehicle, and roadway factors. Similarly, Abdel Aty (2003) used crash data to analyze driver injury severity and showed that roadway characteristics and driver attributes have significant impacts on injury outcomes.

The reports from the studies have established that crashes involving vulnerable road users such as pedestrians, cyclists and motorcyclists often have higher severity of injuries due to lack of physical protection in comparison with vehicle occupants (Eluru et al., 2008). These results suggest the importance of considering multiple variables at the same time when examining crash severity.

### 2.3 Environmental and Road Infrastructure Factors

Environmental conditions are an important factor in traffic safety. Adverse weather conditions such as rain, snow, fog and ice can impair a driver's ability to see, cause a loss in vehicle traction, and increase braking distance. These conditions are often responsible for the increased crash rate and more severe injury rate.

Eisenberg (2004) studied the relationship between precipitation (any kind of weather condition where something's falling from the sky.) and traffic crashes and concluded that rainfall increases the likelihood of accidents as a result of reduced road friction and decreased visibility. Similarly, Qiu and Nixon in 2008 analyzed the effect of adverse weather on traffic crashes and found that adverse weather conditions significantly increase the probability of collisions.

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Road infrastructure characteristics such as road surface conditions, road alignment and intersection design are also critical factors in crash severity. Poor road surface conditions and complicated road alignment can make it hard for drivers to keep their vehicles under control, especially when driving at high speed. Studies have found that curves, slopes and poorly designed intersections are related to higher levels of crash severity (Savolainen et al., 2011).

Traffic control mechanisms - such as traffic signals, stop signs and road markings - are designed to regulate the flow of traffic and to reduce the risk of collisions. However, lack of good traffic control or bad road intersection design could lead to increased chances of serious crashes.

### 2.4 Human and Vehicle Factor

Human characteristics are one of the most important factors in crash severity. Driver demographics, such as age, sex and driving experience, can affect reaction time and risk-taking behavior as well as physical vulnerability in the event of collision. Research has shown that younger drivers tend to exhibit higher levels of risky driving behavior, as well as older drivers being more physically vulnerable in the event of a crash (Abdel Aty, 2003).

Second most important factor affecting crash severity is road user type. Vulnerable road users including pedestrians, cyclists and motorcyclists are more likely to be seriously injured as they do not have the structural protection given by enclosed vehicles. Eluru et al. (2008) found that pedestrian crashes tend to result in significantly higher levels of injury severity compared with vehicle occupant crashes.

Vehicle characteristics also have an effect on crash outcomes. Vehicle type, safety features and vehicle age may influence the amount of protection provided during collisions. Modern vehicles are equipped with serious safety features such as airbags, anti-lock braking

systems, and electronic stability control systems like ADAS, which can reduce the severity of injuries. Older vehicles may not be equipped with advanced safety technologies and hence, offer less safety during crashes (Washington et al., 2011).

### 2.5 Logistic Regression for Crash Severity Analysis

Logistic regression is one of the most popular statistical methods used to analyze crash severity. It is especially useful for modeling binary outcomes like injury/no injury. Logistic regression models are used to estimate the probability of an event occurring given a set of explanatory variables and provide coefficients to identify the direction and strength of relationships between predictors and an outcome.

Lord and Mannering (2010) pointed out that logistic regression models are often used in transportation safety research as they enable the researcher to estimate the effects of multiple variables at the same time. These models also generate odds ratios which can be interpreted as measures of the effect of specific factors on the probability of injury.

If the severity of the crash is in an ordered form, such as minor injury, serious injury, and fatal injury, there is a possibility that the researchers can use ordinal logistic regression models. These models consider the ordered nature of the severity outcomes. However, it has been found that ordinal models can work less well in cases where the categories of severity are very imbalanced, and researchers sometimes choose instead to use binary classification models (Savolainen et al. 2011).

### 2.6 Simulation and Monte Carlo Techniques for Road Safety

In addition to regression analysis, simulation techniques are increasingly used to assess the crash risk under various conditions. Scenario simulation enables researchers to make

estimates of the changes in injury probabilities with a modification of specific variables such as weather conditions, road infrastructure, or traffic control systems.

Sensitivity analysis is often applied to investigate the effect of changes in individual variables on model predictions. By changing one variable at a time and all other factors remain constant, this enables researchers to identify which factors have the greatest effect on the outcome of a crash.

Monte Carlo simulation is another method of analyzing uncertainty in predictive modeling. This method creates thousands of scenarios, based on probability distributions of model variables. The generated simulations are a distribution of predicted outcomes, enabling researchers to assess variability and uncertainty in crash risk predictions (Helbing, 2001).

These types of simulation approaches are complementary to statistical models, as they can add information about the variation of the crash risk under various conditions and scenarios.

## 2.7 Summary of Literature Review

Previous research has shown that crash severity is determined by a combination of environmental conditions, roadway characteristics, human attributes and vehicle-related factors. Logistic regression models have been used extensively to estimate injury probabilities and the effects of these variables. In recent years, simulation methods such as scenario analysis, sensitivity analysis and Monte Carlo simulation have been applied to further investigate crash risk and uncertainty in predictive models.

Although several studies have been conducted on individual factors influencing crash severity, fewer studies have combined predictive statistical modeling with simulation-based approaches using large-scale datasets of accidents. This study fills that gap by combining logistic regression modeling with scenario simulation and Monte Carlo analysis to analyse injury

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severity in road accidents in Canada and to gain insights that can lead to evidence-based transportation safety policies.

## Chapter 3: Data and Methodology

### 3.1 Introduction

This chapter describes the methodology used to analyse injury severity in road accidents in Canada. The study uses statistical modeling and simulation techniques to understand how environmental conditions, road infrastructure and human characteristics, and vehicle attributes affect the likelihood of injury in a traffic collision. The methodology consists of data collection, processing, model development, model evaluation, and simulation-based analysis. Logistic regression models are used to estimate injury probabilities, and scenario simulation and Monte Carlo methods are used to assess how crash risks change with changes to the conditions and to assess uncertainty in model predictions.

### 3.2 Data Description

The dataset used in this study was obtained from the Transport Canada National Collision Database (NCDB) which contains detailed information about traffic collisions reported across Canada. The data set contains variables that describe crash circumstances, weather conditions, road conditions, traffic control measures, human characteristics, vehicle information and the severity of injuries sustained.

The original dataset has about 5.86 million records and 22 variables covering road accidents over multiple years. Each observation consists of a single individual involved in a reported road collision. This dataset contains multiple categories of variables such as environmental factors, roadway characteristics, human attributes and vehicle-related factors.



The large size of the dataset makes it possible to analyze the patterns of crash severity in depth and to help identify factors associated with increased risk of injury using statistical modeling techniques.

### 3.3 Data Preprocessing

Before model development, several preprocessing steps were applied to prepare the dataset for analysis.

#### Data Cleaning

The dataset contains a number of categorical variables that were coded as numerical and alphanumeric values. These variables were standardized and transformed into consistent formats in order to ensure that they are compatible with statistical modeling techniques.

During the process of converting categorical variables to numerical values, some entries were observed to be missing. These entries have been previously recorded as Unknown, choices were other or Jurisdiction Does Not Allow For the purpose of consistency in the analysis, these categories have been combined and classified under one category called 'Other'. This way the study was kept of losing observations without losing too much data.

A significant portion of the dataset contained categorical values such as “U”, “UU”, “X”, “XX”, “N”, and “NN”, which represent unknown, not stated, or unavailable information. Instead of removing these observations, they were grouped into a single category labeled **“Other”**.

This approach was selected for three key reasons:

- **Data Volume Preservation:** Removing all observations with unknown values would lead to a large loss of data, which is especially important since the results of crash severity are already imbalanced.

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- Elimination of Bias: Removal of this type of records may lead to sampling bias, particularly when the unknown's values are correlated with particular crash situations.
- Model Stability: By lumping these values into one category, the model can express the overall effect of the values without creating sparsity in the model due to many low frequencies categories.
- Also, continuous variables were not imputed (PAGE (person age)) and VEHICLEAGE to maintain their original distribution. Time related variables (CMNTH, CWDAY, CHOUR) were also retained in their original economic form since inputting the time related variables may cause distortion in the time series of the crash occurrences.

### Feature Selection

Relevant variables were chosen based on domain knowledge and past research in crash severity analysis, and the variables were divided into two main Groups:

#### Environmental and Road Context Variable

- 1) Weather Conditions (C\_WTHR)
- 2) Road Surface Conditions (C\_RSUR)
- 3) Road Alignment (C\_RALN)
- 4) Traffic Control (C\_TRAF)
- 5) Year(C\_YEAR), Month(C\_MNTH), Day of Week (C\_WDAY), Hour (C\_HOUR)
- 6) Number of Vehicle Involved (C\_VEHS)

#### Human and Vehicle Variables

- 1) Safety Device Used (P\_SAFE)
- 2) Road User Type (P\_USER)
- 3) Sex (P\_SEX)

4) Vehicle Type (V\_TYPE)

5) Vehicle Age (V\_AGE)

6) Person Age (P\_AGE)

### Sampling Strategy

The original dataset contains over 5 million observations, so model estimation can be computationally heavy. To improve efficiency while keeping the statistical representativeness, a random sampling strategy was used.

A sample of about 150,000 observations were randomly selected from the complete sample. This sample size preserves the statistical characteristics of the original data and enables faster model estimation and experimentation during the modeling process.

### 3.4 Train-Test Split

To assess the model performance and prevent overfitting, the data set was split into the training and testing data sets. 70% of the data was used for model training and 30% of the data was used for model testing

The training dataset was used to estimate the model parameters, and the testing dataset was used to evaluate the performance of predictions. This approach helps to ensure that the models are able to generalize to unseen data.

### 3.5 Model Assumption and Diagnostics

The binary and ordinal logistic regression models used in this study rely on several statistical assumptions. While the primary focus of the study was on predictive performance and interpretability, key assumptions were considered during model development.

- Independence of Observations: Each record in the dataset represents an individual involved in a collision. The sampling process ensured that observations were treated independently.

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- **Absence of Perfect Multicollinearity:** To avoid redundancy among predictors, categorical variables were encoded using one-hot encoding with appropriate reference categories. Rare categories were grouped to reduce sparsity and multicollinearity risk.
- **Linearity in the Logit (for continuous variables):** Continuous predictors such as P\_AGE and VEHICLE\_AGE were included in their original form. Their effects were interpreted as incremental changes in the log-odds of injury probability.
- **Large Sample Size Justification:** Given the large dataset (over 100,000 sampled observations), Maximum Likelihood Estimation (MLE) is appropriate and provides stable parameter estimates.
- **Class Imbalance Consideration:** The initial modeling using collision-level severity revealed extreme imbalance in fatal outcomes. This issue was addressed by shifting to person-level injury severity and later simplifying the outcome into a binary classification to improve model stability.

### 3.6 Summary of Data and Methodology

This chapter presented the data set and analytical method which was used to study crash injury severity in Canadian road accidents. The study utilized the National Collision Database (NCDB) from Transport Canada which had over 5.8 million records of car accidents with variables related to environmental conditions, road characteristics, human factors and vehicle attributes. Data preprocessing steps such as cleaning, rare categories grouping and features selection were applied to prepare dataset for analysis. A random sample of 150,000 observations was used to make computational processes more efficient, and the dataset was split into training and testing datasets (70:30).

## Chapter 4: Model Development

### 4.1 Model Development Process

During the modeling phase, various statistical approaches were tested to identify the most suitable approach for analyzing the crash severity in the dataset. Due to the large size of the data set and multiple severity categories, various modeling approaches were explored before choosing the final approach.

Initially, this study tried to model crash severity based on the C\_SEV variable which represents different injury severity categories, but during exploratory analysis, it was observed that there is severe class imbalance in the dataset, where non injury cases are the major part of dataset and fatal injury cases represent very small percentage of all observations, making it difficult for model to accurately learn the patterns associated with rare events like fatal crashes

To resolve this issue, the study next tried to use an ordinal logistic regression model using the P\_ISEV variable, in which the outcome of injury was categorized into three groups: no injury, injury, and fatal injury. Ordinal regression was selected because the levels of severity follow a natural order of ranking. However, during model estimation it was noted that the number of observations for fatal injury was extremely low compared to the other categories. As a result, the model found it difficult to correctly predict cases of fatal injuries, resulting in unstable estimates and poor predictive performance for the highest severity category of injuries.

To increase the stability of the model, the injury categories injury and fatal injury were grouped into a single category, with the resulting model being a binary outcome of injury versus no injury. This transformation enabled the problem to be modelled with the binary logistic regression, which is more robust when dealing with imbalanced severity distribution.

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Binary logistic regression models were then estimated using two different configurations:

1. Model with grouped "Other" categories
2. Model without grouped "Other" categories

When the model included an "Other" category, it allowed rare or unknown values to be grouped together instead of being removed from the dataset. During the analysis process, it was noted that the "Other" group seemed to play a strong role in the probability of injury, which required further investigation. In order to determine if the effect was caused by aggregation of data, a second model was established without including "Other" group. However, dropping the grouped category decreased the model stability and resulted in lower overall predictive accuracy for the final model which was for this reason the grouped category approach was retained.

To better understand the influence of types of variables, the analysis was split into two separate research question models:

### RQ1 – Environmental and Road Context Model

This model included variables related to environmental and roadway conditions, such as:

- 1) Weather conditions
- 2) Road surface condition
- 3) Road alignment
- 4) Traffic control systems
- 5) Time-related variables

### RQ2 – Human and Vehicle Model

This model focused on factors related to road users and vehicles, including:

- 1) Safety device usage
- 2) Road user type

- 3) Sex
- 4) Vehicle type
- 5) Vehicle age
- 6) Person age

After estimating these two models separately, a combined one was developed which combined both environmental and human-vehicle variables. The performance of all the models was then compared using multiple evaluation metrics such as accuracy, balanced accuracy, confusion matrix, macro F1 score and ROC-AUC.

The combined model performed best in terms of predictive performance and gave a more complete understanding of crash severity by including multiple domains of crash risk.

### 4.2 Ordinal Logistic Regression

Ordinal logistic regression was initially considered because the injury severity variable has ordered categories: no injury, injury, and fatal injury. Using this modeling approach study can estimate the probability of being in higher severity while considering ordered nature of the outcome variable.

However, during model estimation it was observed that the number of fatal injury observations was extremely small compared with the other categories of severity. This imbalance resulted in instability in the ordinal regression model and limited its ability to accurately predict fatal injury outcomes.

To increase the stability of the model and the predictive performance, the injury and fatal injury categories were merged into single categories representing injury cases and thus the severity variable was transformed into a binary outcome variable which then used in a binary logistic regression for final analysis.

### 4.3 Logistic Regression Model

Binary logistic regression was used to estimate the probability of injury occurrence in Traffic accidents. Logistic regression is suitable in the case of a binary dependent variable.

The model coefficients were estimated by using Maximum Likelihood Estimation (MLE). MLE finds values of parameters that maximizes the probability of observing the given dataset.

The estimated coefficients were converted into odds ratios in order to interpret the impact of each variable on injury probability.

### 4.4 Model Evaluation Metrics

Several evaluation metrics were applied to evaluate the performance of the model:

- Accuracy : Measures the percentage of correct predictions.
- Balanced Accuracy: Accounts for the problem of class imbalance by taking the average recall of the classes.
- Precision: Measures the number of predicted cases of injury that are actually correct.
- Recall: Measures of how many actual cases of injury were correctly identified.
- F1 Score: Represents the harmonic average of precision and recall.
- ROC-AUC: Measures the ability of the model to discriminate between injured and non-injured cases, for various classification thresholds.
- Confusion Matrix: Used to summarize the classification results in terms of the actual outcomes versus the predicted outcomes.



### 4.5 Summary of Model Development

This chapter described the modeling strategy applied to examining crash injury severity. Because of extreme class imbalance in original severity categories, ordinal logistic regression provided unstable estimates. To enhance the stability of the models, categories of injury and fatal injury were merged to create a binary outcome of injury and no injury. Binary logistic regression models were then developed for the environmental, road, traffic, human, vehicle factors, and a combined set of variables. Model performance was evaluated by taking metrics such as accuracy, balanced accuracy, F1 score, ROC-AUC and confusion matrix. The best predictive performance was obtained using the combined model, suggesting the severity of a crash is affected by a number of interacting factors.

## Chapter 5: Results and Findings

### 5.1 Introduction

This chapter provides the results of the statistical models and simulation analyses that were used to examine injury severity in Canadian road accidents. The results are presented in accordance with the research questions. First, the outcomes of the environmental and traffic context model (RQ1) are presented. Second, the results of the human and vehicle model (RQ2) are covered. Third, the results of the combined model are presented and compared to the individual models.

### 5.2 Binary Logistic Regression

#### 5.2.1 RQ1: Environmental, Road and Traffic Context Model

The binary logistic regression model for RQ1 examined the effect of weather, road surface, road alignment, traffic control, month, weekday, hour, and number of vehicles involved on injury occurrence.

Results of RQ1 (with other group)

Model Performance

- Accuracy:- 0.5777
- Balanced Accuracy:- 0.5498
- Macro Recall:- 0.5498
- Macro F1:- 0.5414
- ROC AUC:- 0.6012

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Table 5.1: Binary RQ1 With Other Group Confusion Matrix

| Actual / Predicted | No Injury | Injury |
|--------------------|-----------|--------|
| No Injury          | 3300      | 6335   |
| Injury             | 3071      | 9568   |

Table 5.2: Binary RQ1 With Other Group Top Significant Predictors Increasing Injury Probability

| Variable                            | Odds Ratio |
|-------------------------------------|------------|
| C_RSUR_6(Sand/Gravel/Dirt)          | 2.34       |
| C_RALN_6(Bottom of hill)            | 1.92       |
| C_TRAF_02(Traffic signals flashing) | 1.79       |
| C_RALN_3(Curved and level)          | 1.59       |
| C_RALN_4(Curved with gradient)      | 1.58       |
| C_RSUR_7(Muddy)                     | 1.52       |
| C_WTHR_7(Strong Wind)               | 1.49       |

Table 5.3: Binary RQ1 With Other Group Top Significant Predictors Decreasing Injury Probability

| Variable                 | Odds Ratio |
|--------------------------|------------|
| C_VEHS (No of Vehicle)   | 0.76       |
| C_WTHR_3(Bottom of hill) | 0.89       |

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| Variable                                 | Odds Ratio |
|------------------------------------------|------------|
| C_RALN_OTHER(Road Alignment Other Group) | 0.96       |

Results of RQ1 (without other group)

Model Performance

- Accuracy:- 0.5739
- Balanced Accuracy:- 0.5451
- Macro Recall:- 0.5451
- Macro F1:- 0.5361
- ROC AUC:- 0.5988

Table 5.4: Binary RQ1 Without Other Group Confusion Matrix

| Actual / Predicted | No Injury | Injury |
|--------------------|-----------|--------|
| No Injury          | 2757      | 5495   |
| Injury             | 2650      | 8213   |

Table 5.5: Binary RQ1 Without Other Group Top Significant Predictors Increasing Injury

Probability

| Variable                   | Odds Ratio |
|----------------------------|------------|
| C_RSUR_6(Sand/Gravel/Dirt) | 2.25       |
| C_RALN_6(Bottom of hill)   | 2.25       |
| C_RSUR_7(Muddy)            | 1.86       |

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| Variable                       | Odds Ratio |
|--------------------------------|------------|
| C_RALN_3(Curved and level)     | 1.58       |
| C_RALN_4(Curved with gradient) | 1.54       |
| C_WTHR_6(Limited Visibility)   | 1.52       |
| C_TRAF_18(No traffic Control)  | 1.48       |

Table 5.6: Binary RQ1 Without Other Group Top Significant Predictors Decreasing Injury Probability

| Variable                       | Odds Ratio |
|--------------------------------|------------|
| C_VEHS (No of Vehicle)         | 0.77       |
| C_WTHR_3(Bottom of hill)       | 0.91       |
| C_TRAF_6(Pedestrian crosswalk) | 0.97       |

### Summary of RQ1 Model

The findings reveal that RQ1 binary logistic regression model has moderate predictive characteristics with value of ROC-AUC approximately 0.60. The model with grouped "Other" categories represented slightly better as compared to that of the model without grouped categories.

In both model specifications, the strongest predictors that were linked with high probability of injuries were connected with the risky road surface conditions and complicated road alignment features, such as sand or gravel surface, muddy conditions of the road, and

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curved road segments. There were also certain conditions of traffic control like flashing signals or lack of traffic control equipment that was linked with increased probability of injuries.

However, C\_VEHS (No of Vehicle) displayed an odds ratio that was always below 1 in both versions of the model, thus showing that the more vehicles involved, the less likely they are to be injured.

### 5.2.2 RQ2: Human and Vehicle Model

The binary logistic regression model for RQ2 examined the effects of safety device use, road-user type, sex, vehicle type, vehicle age, and person age.

Results of RQ2 with other group

Model Performance

- Accuracy:- 0.6366
- Balanced Accuracy:- 0.6178
- Macro Recall:- 0.6178
- Macro F1:- 0.6177
- ROC AUC:- 0.6894

Table 5.7: Binary RQ2 With Other Group Confusion Matrix

| Actual / Predicted | No Injury | Injury |
|--------------------|-----------|--------|
| No Injury          | 4656      | 5065   |
| Injury             | 3111      | 9668   |

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Table 5.8: Binary RQ2 With Other Group Top Significant Predictors Increasing Injury Probability

| Variable                             | Odds Ratio |
|--------------------------------------|------------|
| P_USER_3 (Pedestrian)                | 230.38     |
| P_USER_4 (Bicyclist)                 | 31.59      |
| P_USER_5(Motorcyclist)               | 12.26      |
| P_SAFE_12 (Other Safety Devices)     | 5.49       |
| P_USER_OTHER (Road User Class Other) | 4.41       |

Table 5.9: Binary RQ2 With Other Group Top Significant Predictors Decreasing Injury Probability

| Variable                                | Odds Ratio |
|-----------------------------------------|------------|
| P_SEX_OTHER(Person Sex Other Group)     | 0.25       |
| V_TYPE_09(School Bus)                   | 0.26       |
| V_TYPE_OTHER(Vehicle Type Other Group)  | 0.32       |
| P_SAFE_OTHER(Safety Device Other Group) | 0.35       |
| V_TYPE_11(Urban and Intercity Bus)      | 0.53       |
| P_SEX_M(Male Drivers)                   | 0.56       |

Results of RQ2 (without other group)

Model Performance

- Accuracy:- 0.6021

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- Balanced Accuracy:- 0.5687
- Macro Recall:- 0.5687
- Macro F1:- 0.5641
- ROC AUC:- 0.622

Table 5.10: Binary RQ2 Without Other Group Confusion Matrix

| Actual / Predicted | No Injury | Injury |
|--------------------|-----------|--------|
| No Injury          | 2799      | 4804   |
| Injury             | 2457      | 8189   |

Table 5.11: Binary RQ2 Without Other Group Top Significant Predictors Increasing Injury

Probability

| Variable                                 | Odds Ratio |
|------------------------------------------|------------|
| P_USER_4 (Bicyclist)                     | 2.25       |
| P_SAFE_12 (Other Safety Devices)         | 2.25       |
| P_SAFE_09 (Helmet Worn)                  | 1.86       |
| P_USER_5(Motorcyclist)                   | 1.58       |
| P_SAFE_13(No Safety Device)              | 1.54       |
| VEHICLE_AGE_MISSING(Missing Vehicle Age) | 1.52       |
| P_AGE_MISSING(Missing Person Age)        | 1.48       |



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Table 5.12: Binary RQ2 Without Other Group Top Significant Predictors Decreasing Injury Probability

| Variable                           | Odds Ratio |
|------------------------------------|------------|
| V_TYPE_09(School Bus)              | 0.37       |
| V_TYPE_08(Tractor)                 | 0.47       |
| V_TYPE_11(Urban and Intercity Bus) | 0.49       |
| V_TYPE_05(Cargo Van)               | 0.52       |
| P_SEX_M(Male Drivers)              | 0.55       |

### Summary of RQ2 Model

The binary logistic regression model (RQ2) has a better predictive ability than the environmental model with ROC-AUC values of between 0.62 to 0.69. The model that groups the categories of Other performs better than the non-grouping model. In both model specifications, the most influential factor is road user type, and odds of injury are considerably greater among the vulnerable road users, including pedestrians, bicyclists, and motorcyclists. Other important predictors are also identified as safety-related and vehicle-related variables. On the whole, the model shows that the combination of human and vehicle factors is a significant factor in the injury outcome.

The positive association of helmet use with injury probability reflects exposure bias, as helmet use is more common among high-risk road users such as motorcyclists.

### 5.2.3 Combined model

The combined binary logistic regression model incorporated both environmental variables from RQ1 and human and vehicle-related variables from RQ2 in order to evaluate whether integrating all predictors improves injury severity prediction.

Results of Combined Model with other group

Model Performance

- Accuracy:- 0.6644
- Balanced Accuracy:- 0.6502
- Macro Recall:- 0.6502
- Macro F1:- 0.6514
- ROC AUC:- 0.7255

Table 5.13: Binary Combined With Other Group Confusion Matrix

| Actual / Predicted | No Injury | Injury |
|--------------------|-----------|--------|
| No Injury          | 5250      | 4385   |
| Injury             | 3090      | 9549   |

Table 5.14: Binary Combined With Other Group Top Significant Predictors Increasing Injury

Probability

| Variable               | Odds Ratio |
|------------------------|------------|
| P_USER_3 (Pedestrian)  | 180.39     |
| P_USER_4 (Bicyclist)   | 46.93      |
| P_USER_5(Motorcyclist) | 11.30      |

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| Variable                             | Odds Ratio |
|--------------------------------------|------------|
| P_SAFE_12 (Other Safety Devices)     | 3.63       |
| P_USER_OTHER (Road User Class Other) | 2.83       |
| C_RSUR_6(Sand/Gravel/Dirt)           | 2.64       |
| P_SAFE_09 (Helmet Worn)              | 2.17       |
| C_RSUR_7(Muddy)                      | 2.11       |
| C_RALN_6(Bottom of Hill)             | 1.95       |
| C_RALN_3(Curved and level)           | 1.75       |

Table 5.15: Binary Combined With Other Group Top Significant Predictors Decreasing Injury Probability

| Variable                                | Odds Ratio |
|-----------------------------------------|------------|
| P_SEX_OTHER(Person Sex Other Group)     | 0.22       |
| V_TYPE_09(School Bus)                   | 0.29       |
| P_SAFE_OTHER(Safety Device Other Group) | 0.30       |
| V_TYPE_OTHER(Vehicle Type Other Group)  | 0.42       |
| V_TYPE_07(Unit Truck)                   | 0.47       |
| V_TYPE_08(Road Tractor)                 | 0.48       |
| V_TYPE_11(Urban and Intercity Bus)      | 0.53       |
| P_SEX_M(Male Drivers)                   | 0.55       |

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| Variable             | Odds Ratio |
|----------------------|------------|
| V_TYPE_05(Cargo Van) | 0.56       |

Results of Combined Model (without other group)

Model Performance

- Accuracy:- 0.6391
- Balanced Accuracy:- 0.6097
- Macro Recall:- 0.6097
- Macro F1:- 0.60992
- ROC AUC:- 0.6772

Table 5.16: Binary Combined Without Other Group Confusion Matrix

| Actual / Predicted | No Injury | Injury |
|--------------------|-----------|--------|
| No Injury          | 2812      | 3632   |
| Injury             | 1968      | 7103   |

Table 5.17: Binary Combined Without Other Group Top Significant Predictors Increasing Injury

Probability

| Variable                   | Odds Ratio |
|----------------------------|------------|
| P_USER_4 (Bicyclist)       | 33.05      |
| P_USER_5(Motorcyclist)     | 4.12       |
| C_RSUR_6(Sand/Gravel/Dirt) | 3.97       |

## COLLISION SEVERITY ANALYSIS IN CANADIAN ROAD ACCIDENTS

| Variable                         | Odds Ratio |
|----------------------------------|------------|
| P_SAFE_09 (Helmet Worn)          | 3.37       |
| P_SAFE_12 (Other Safety Devices) | 3.37       |
| C_RSUR_7(Muddy)                  | 2.54       |
| C_RALN_6(Bottom of Hill)         | 1.95       |

Table 5.18: Binary Combined Without Other Group Top Significant Predictors Decreasing Injury Probability

| Variable                           | Odds Ratio |
|------------------------------------|------------|
| V_TYPE_09(School Bus)              | 0.29       |
| V_TYPE_08(Tractor)                 | 0.39       |
| V_TYPE_11(Urban and Intercity Bus) | 0.48       |
| V_TYPE_07(Unit Truck)              | 0.51       |
| P_SEX_M(Male Drivers)              | 0.53       |
| V_TYPE_05(Cargo Van)               | 0.55       |

### Summary of Combined Model

Binary logistic regression model combined has the highest predictive value of all models and the ROC-AUC scores of the model with grouped categories of Other are more than 0.72.

The findings show that the model predicts injury outcomes better in case when environmental, road, human, and vehicle-related variables integrate. The most effective predictors are those related to road condition, as well as those related to road users, which indicates that the severity of crashes is affected by a set of interdependent factors.. The model in which the Other categories are grouped has a higher overall performance than the model in which the categories were eliminated.

The positive association of helmet use with injury probability reflects exposure bias, as helmet use is more common among high-risk road users such as motorcyclists.

### 5.3 Ordinal Regression Model

#### 5.3.1 RQ1: Environmental, Road and Traffic Context Model

The Ordinal regression model for RQ1 examined the effect of weather, road surface, road alignment, traffic control, month, weekday, hour, and number of vehicles involved on injury occurrence.

Results of RQ1 (with other group)

Model Performance

- Accuracy:- 0.5779
- Balanced Accuracy:- 0.3598
- Macro Recall:- 0.3598
- Macro F1:- 0.3339
- ROC AUC:- 0.46334
- MAE:- 0.4229

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Table 5.19: Ordinal RQ1 With Other Group Confusion Matrix

| Actual / Predicted | No Injury | Injury | Fatal Injury |
|--------------------|-----------|--------|--------------|
| No Injury          | 2018      | 7618   | 0            |
| Injury             | 1623      | 10854  | 0            |
| Fatal Injury       | 17        | 144    | 0            |

Results of RQ1 (without other group)

Model Performance

- Accuracy:- 0.582
- Balanced Accuracy:- 0.3638
- Macro Recall:- 0.3638
- Macro F1:- 0.3423
- ROC AUC:- 0.6286
- MAE:- 0.4186

Table 5.20: Ordinal RQ1 Without Other Group Confusion Matrix

| Actual / Predicted | No Injury | Injury | Fatal Injury |
|--------------------|-----------|--------|--------------|
| No Injury          | 1932      | 6320   | 0            |
| Injury             | 1529      | 9193   | 0            |
| Fatal Injury       | 12        | 129    | 0            |

### Summary of RQ1 Ordinal Model

The ordinal regression model of RQ1 indicates moderate overall accuracy, but low balanced accuracy and macro F1, which reflect constraints in predicting at all levels of severity. The values of ROC-AUC also differ among model specifications, as the model with no grouped categories shows a slight better result.

The results of the confusion matrix have shown that the model mostly predicts no injury and injury whereas no successful prediction of fatal injury cases is made. This indicates the ordinal model gap in dealing with extremely unbalanced severity classes.

### 5.3.2 RQ2: Human and Vehicle Model

The Ordinal regression model for RQ2 examined the effects of safety device use, road-user type, sex, vehicle type, vehicle age, and person age.

#### Results of RQ2 with other group

##### Model Performance

- Accuracy:- 0.5761
- Balanced Accuracy:- 0.3815
- Macro Recall:- 0.3815
- Macro F1:- 0.3804
- ROC AUC:- 0.5716
- MAE:- 0.4266

Table 5.21: Ordinal RQ2 With Other Group Confusion Matrix

| Actual / Predicted | No Injury | Injury | Fatal Injury |
|--------------------|-----------|--------|--------------|
| No Injury          | 4364      | 4347   | 0            |



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| Actual / Predicted | No Injury | Injury | Fatal Injury |
|--------------------|-----------|--------|--------------|
| Injury             | 3838      | 6929   | 0            |
| Fatal Injury       | 52        | 73     | 0            |

Results of RQ2 (without other group)

Model Performance

- Accuracy:- 0.5877
- Balanced Accuracy:- 0.3802
- Macro Recall:- 0.3802
- Macro F1:- 0.3782
- ROC AUC:- 0.6009
- MAE:- 0.4134

Table 5.22: Ordinal RQ2 Without Other Group Confusion Matrix

| Actual / Predicted | No Injury | Injury | Fatal Injury |
|--------------------|-----------|--------|--------------|
| No Injury          | 3122      | 4189   | 0            |
| Injury             | 2823      | 7028   | 0            |
| Fatal Injury       | 19        | 89     | 0            |

Summary of RQ2 Ordinal Model

ROC-AUC values of between 0.57 and 0.60 show that the RQ2 ordinal regression model has mediocre predictive performance. The model without grouped categories appears to be slightly more accurate than one with grouping.

Nevertheless, just like the RQ1 ordinal model, the confusion matrix shows that the model does not predict the fatal injury cases, with the predictions concentrated in the lower severity categories. This implies that ordinal model fails to effectively render rare extreme results in the data.

### 5.3.3 Combined model

The combined Ordinal regression model incorporated both environmental variables from RQ1 and human and vehicle-related variables from RQ2 in order to evaluate whether integrating all predictors improves injury severity prediction.

Results of Combined Model with other group

Model Performance

- Accuracy:- 0.5999
- Balanced Accuracy:- 0.3884
- Macro Recall:- 0.3884
- Macro F1:- 0.3816
- ROC AUC:- 0.6463
- MAE:-0.4009

Table 5.23: Ordinal Combined With Other Group Confusion Matrix

| Actual / Predicted | No Injury | Injury | Fatal Injury |
|--------------------|-----------|--------|--------------|
| No Injury          | 3284      | 5353   | 0            |
| Injury             | 2292      | 8366   | 0            |
| Fatal Injury       | 17        | 107    | 0            |

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### Results of Combined Model (without other group)

#### Model Performance

- Accuracy:- 0.6117
- Balanced Accuracy:- 0.3828
- Macro Recall:- 0.3828
- Macro F1:- 0.3712
- ROC AUC:- 0.6469
- MAE:- 0.3893

Table 5.24: Ordinal Combined Without Other Group Confusion Matrix

| Actual / Predicted | No Injury | Injury | Fatal Injury |
|--------------------|-----------|--------|--------------|
| No Injury          | 1868      | 4320   | 0            |
| Injury             | 1291      | 7122   | 0            |
| Fatal Injury       | 15        | 80     | 0            |

#### Summary of Combined Ordinal Model

The combined ordinal regression model demonstrates a better performance than the isolated ordinal models with the ROC-AUC values of approximately 0.64. The non-grouped category model has a slightly better accuracy and reduced prediction error (MAE).

In spite of this, the model still has limitations in predicting cases of fatal injury as resulted in the confusion matrix. Results show that combination of variables improves performance of a model, but the ordinal method is limited by the imbalance in classes of the severity categories.

## 5.4 Model Comparison

Table 5.25: All Model Comparison

| Model                  | Configuration | Accuracy      | Balanced Accuracy | Macro F1      | ROC-AUC       |
|------------------------|---------------|---------------|-------------------|---------------|---------------|
| RQ1 Binary Model       | With Other    | 0.5777        | 0.5498            | 0.5414        | 0.6012        |
| RQ1 Binary Model       | Without Other | 0.5739        | 0.5451            | 0.5361        | 0.5988        |
| RQ2 Binary Model       | With Other    | 0.6366        | 0.6178            | 0.6177        | 0.6894        |
| RQ2 Binary Model       | Without Other | 0.6021        | 0.5687            | 0.5641        | 0.6220        |
| Combined Binary Model  | With Other    | <b>0.6644</b> | <b>0.6502</b>     | <b>0.6514</b> | <b>0.7255</b> |
| Combined Binary Model  | Without Other | 0.6391        | 0.6097            | 0.6099        | 0.6772        |
| RQ1 Ordinal Model      | With Other    | 0.5779        | 0.3598            | 0.3339        | 0.4633*       |
| RQ1 Ordinal Model      | Without Other | 0.5820        | 0.3638            | 0.3423        | 0.6286        |
| RQ2 Ordinal Model      | With Other    | 0.5761        | 0.3815            | 0.3804        | 0.5716        |
| RQ2 Ordinal Model      | Without Other | 0.5877        | 0.3802            | 0.3782        | 0.6009        |
| Combined Ordinal Model | With Other    | 0.5999        | 0.3884            | 0.3816        | 0.6463        |
| Combined Ordinal Model | Without Other | 0.6117        | 0.3828            | 0.3712        | 0.6469        |

### Summary of Model Comparison

Comparison of binary and ordinal regression reveals that binary logistic regression is more important in predicting crash severity. Binary models are always more accurate, balanced accurate, and have greater F1 scores and ROC-AUC than ordinal models.

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The combined binary logistic regression model with grouped Other categories is the best, however, the results suggest that the combination of the environmental, road, human and vehicle related factors enhances the predictive ability.

Conversely, ordinal regression models demonstrate weaknesses with regard to the skewed nature of the severity levels especially in prediction of fatal cases of injury. These results imply that the most appropriate model to use in this data is binary classification to model crash injury severity.

## Chapter 6: Simulation and Monte Carlo Analysis

### 6.1 Introduction

Simulation-based analysis is introduced in this chapter to assess the variations in the probability of injuries in varying crash conditions. Although regression models define the existence of significant predictors, the simulation methods permit the evaluation of the impact of these factors on the injury risk in the real-life settings.

The combined binary logistic regression model which showed the best predictive power is used to apply scenario simulation, sensitivity analysis and Monte Carlo simulation.

### 6.2 Scenario Simulation

A baseline crash profile under normal conditions, i.e. clear weather, dry road, straight road, traffic light control and standard vehicle occupant, was used to perform a scenario simulation. The predicted probability of injury in the base case is 0.5479 (95% CI: 0.5390 to 0.5568).

The effects of variations in the probability of injuries were compared by varying the individual variables:

- The elimination of traffic signal control increased the risk of injury by 17.11%.
- Changing weather from normal to severe increased probability by 15.76%
- Risky road surface conditions increased probability by 36.46%
- Curved road alignment increased probability by 25.53%
- Changing road user type to motorcyclist increased probability by 68.62%

Findings indicate that the probability of injury continues to rise significantly when conditions of high risks are added, especially when it comes to vulnerable road users and poor road conditions.

### 6.3 Sensitivity Analysis

A sensitivity analysis was conducted to determine how each variable affects the probability of injury.

The result of Sensitivity Analysis indicated:

- Changing sex from female driver to male driver reduces predicted injury probability by approximately 28.12%
- Vehicle age shows a gradual increase in injury probability as vehicle age increases
- Person age also shows a steady increase in injury probability with increasing age

These findings confirm that demographic and vehicle-related variables have an effect on injury risk, but their effects are less significant than those of road user type and road conditions.

### 6.4 Monte Carlo Simulation Analysis

In order to determine the variability in injury probability, Monte Carlo simulation was undertaken with 5,000 randomly generated scenarios.

The baseline probability of injury is 0.5479 and simulation results indicate:

- Mean probability: 0.5462
- Median probability: 0.5460
- Standard deviation: 0.1593
- 5th percentile: 0.2860
- 95th percentile: 0.8139

About 9.6% of simulated cases showed a high probability of injury (more than or equal to 0.75).

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Risk band distribution shows:

- 3.3% of cases below 0.25
- 35.36% between 0.25 and 0.50
- 51.70% between 0.50 and 0.75
- 9.64% above 0.75

These findings demonstrate a high level of variability in the risk of injury based on the crash conditions.

### 6.5 Summary of simulation analysis

The outcomes of the simulation have shown that the probability of injuries is very sensitive to the variations in key variables, especially road user type and road conditions. The scenario analysis reveals that the interaction between various risk factors produces significant changes in the probability of injuries.

Monte Carlo simulation emphasizes the uncertainty in the risk of injury through various situations, with a small percentage of cases having very high risk. These results underline the significance of the multiple interacting factors in the analysis of the severity of crashes.



## Chapter 7: Data Visualization

### 7.1 Model Performance Comparison

The performance of different models is compared using balanced accuracy and ROC-AUC.

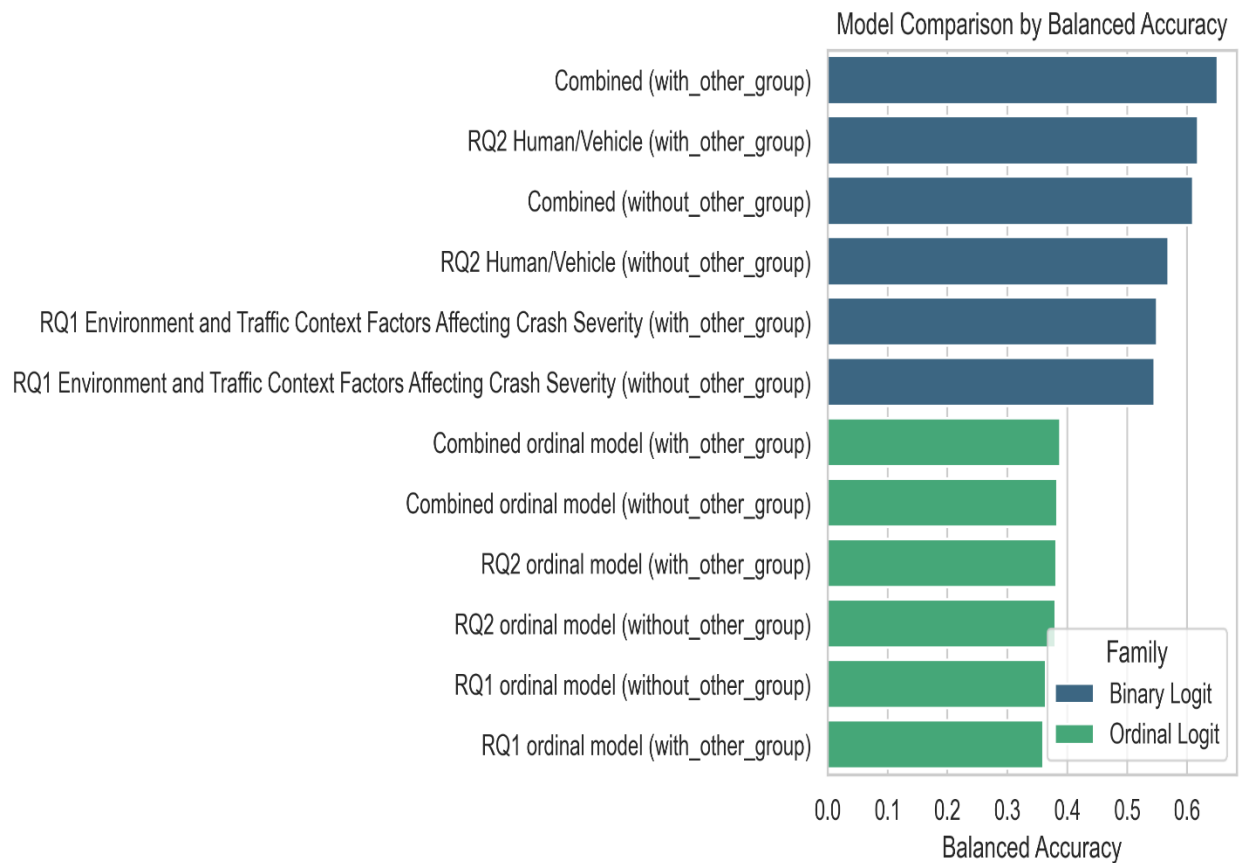


Figure 1: Model Comparison – Balanced Accuracy

This figure compares the balanced accuracy of different models, showing the relative performance of binary and ordinal approaches.

## COLLISION SEVERITY ANALYSIS IN CANADIAN ROAD ACCIDENTS

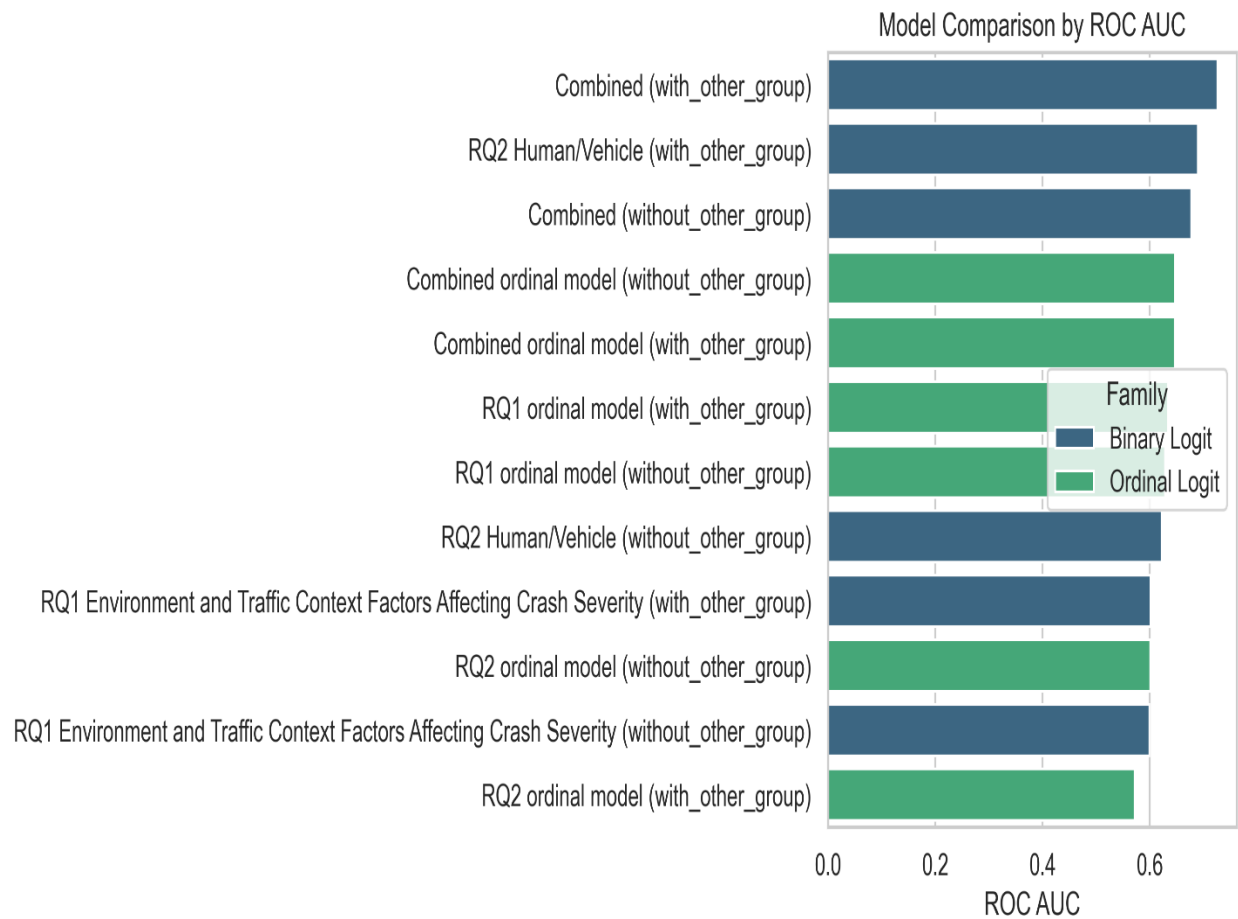


Figure 2: Model Comparison – ROC AUC

This figure illustrates the ROC-AUC values for all models, highlighting differences in predictive performance.

## 7.2 Visualization of Key Predictors

The most influential variables were visualized using odds ratios.

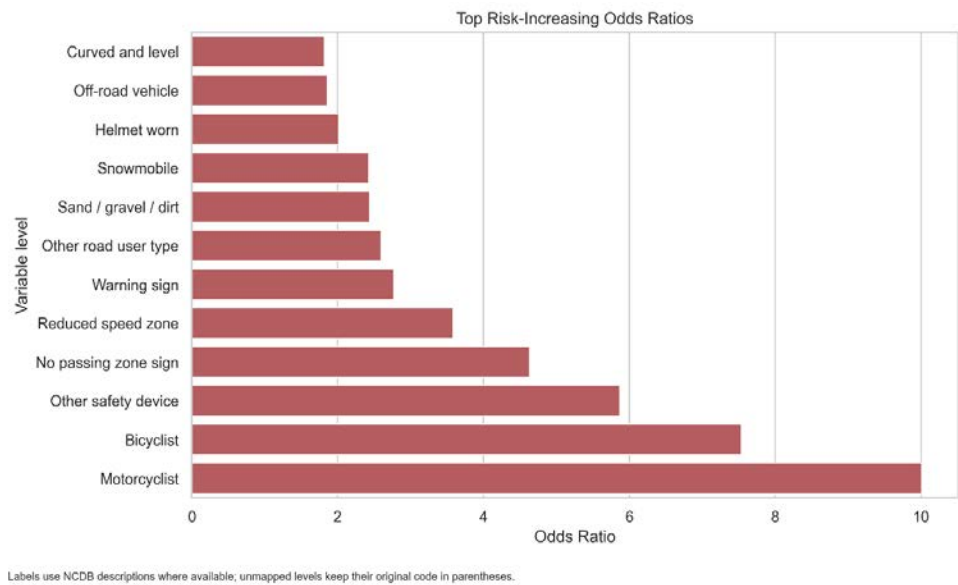


Figure 3: Top Risk-Increasing Factors

This figure displays the variables associated with the highest increase in injury probability.

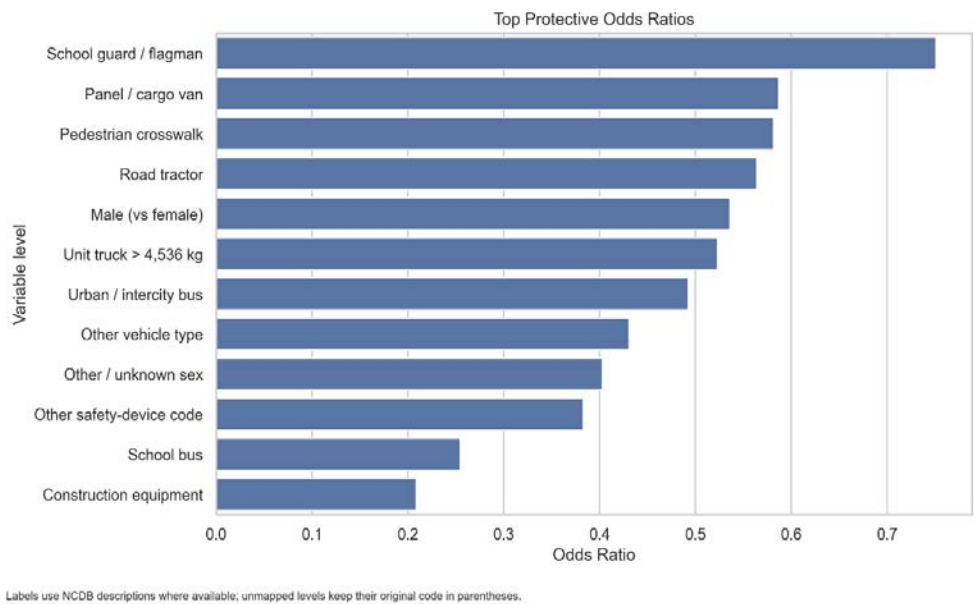


Figure 4: Top Protective Factors

This figure presents the variables associated with reduced injury probability.

7.3 Scenario and Sensitivity Visualization

Visualizations are used to understand how injury probability changes under different conditions.

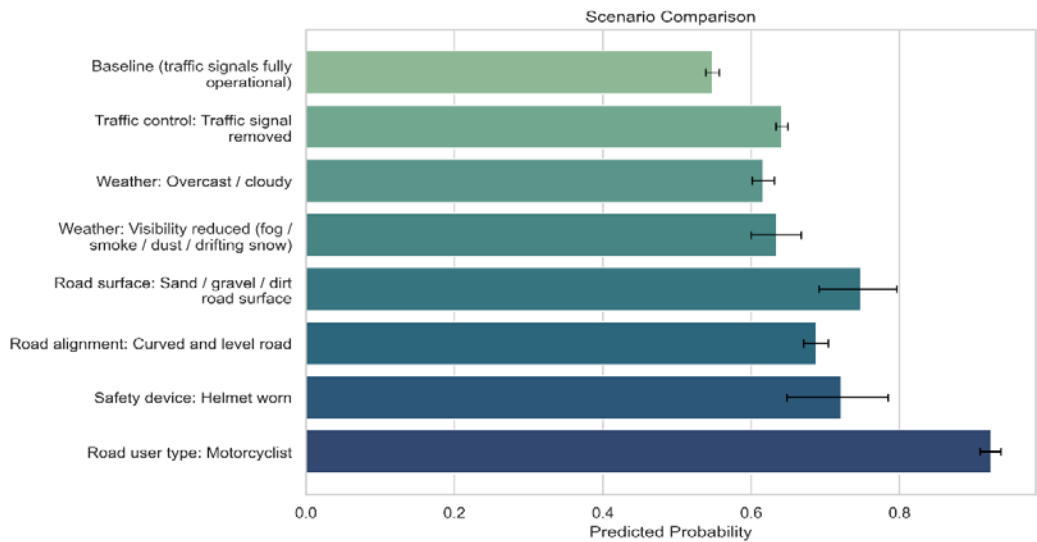


Figure 5: Scenario Comparison Results

This figure illustrates how predicted injury probability changes under different condition

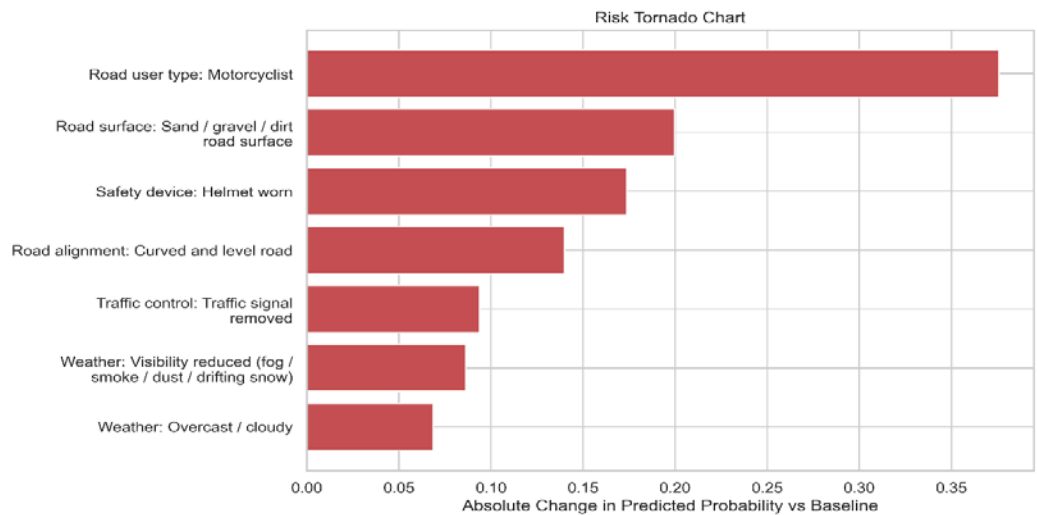


Figure 6: Risk Tornado Chart

This figure shows the relative impact of key variables on injury probability, ranked by their influence.

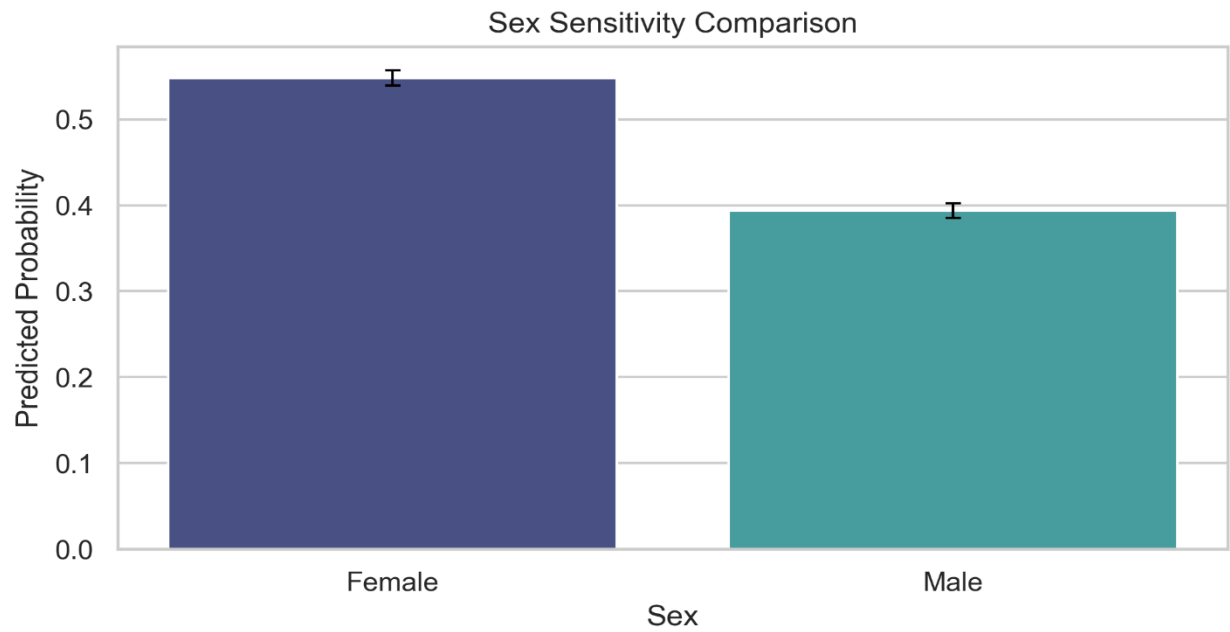


Figure 7: Sex Sensitivity Comparison

This figure compares predicted injury probability between male and female drivers.

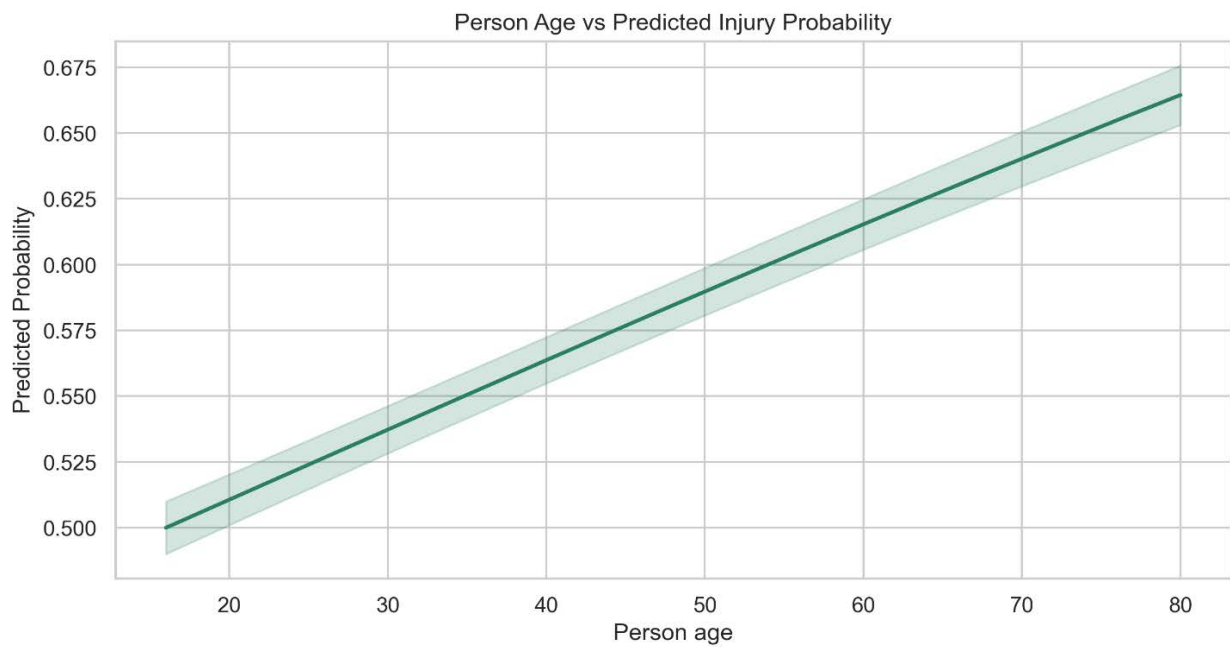


Figure 8: Person Age vs Predicted Probability

This figure shows how predicted injury probability changes with increasing person age.

7.4 Monte Carlo Simulation Visualization

Monte Carlo simulation results are visualized to show distribution and variability of risk.

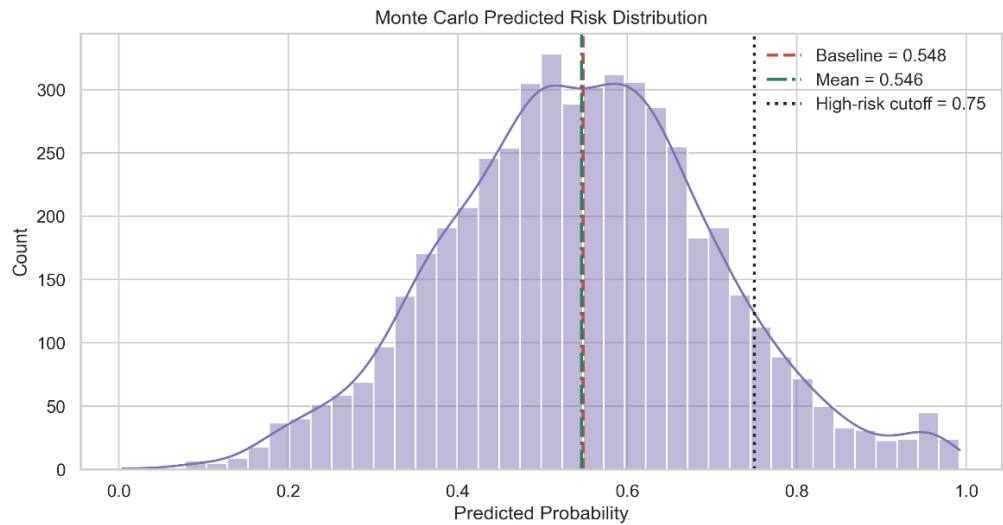


Figure 9: Monte Carlo Risk Distribution

This figure presents the distribution of predicted injury probabilities generated through Monte Carlo simulation.

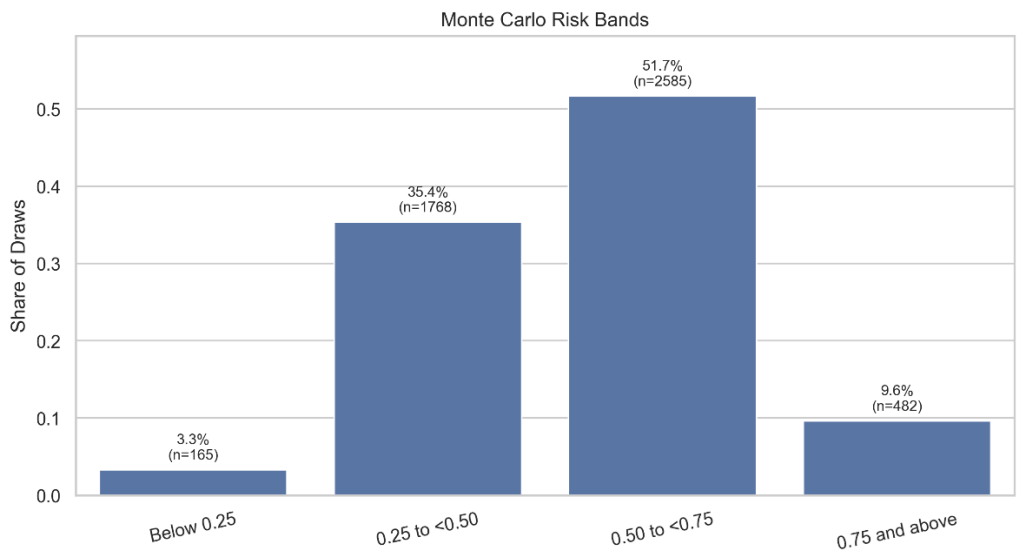


Figure 10: Monte Carlo Risk Bands

This figure shows the proportion of simulated scenarios across different risk probability ranges.

## COLLISION SEVERITY ANALYSIS IN CANADIAN ROAD ACCIDENTS

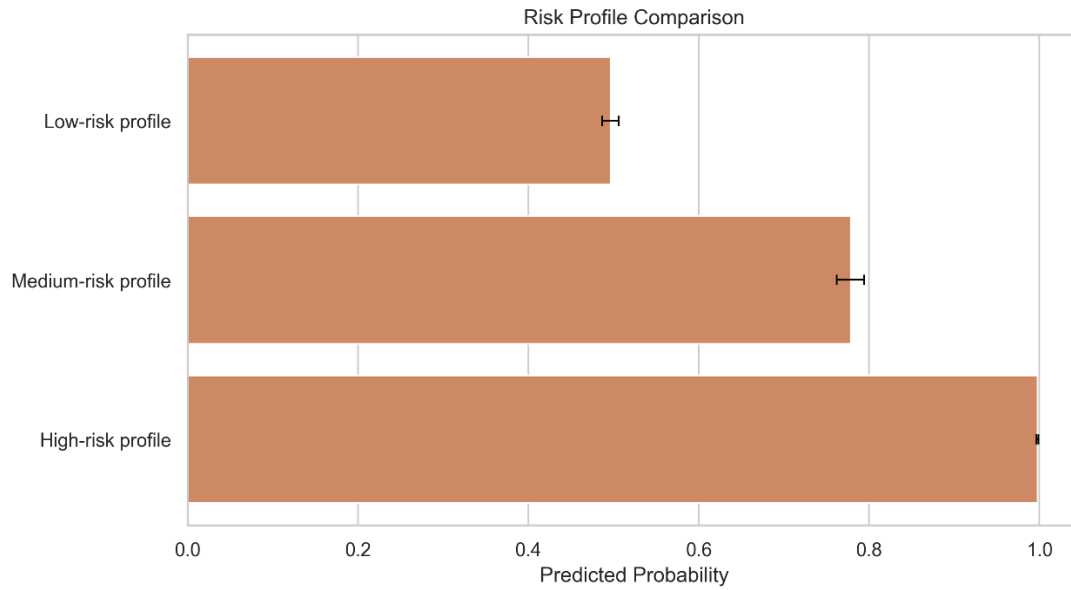


Figure 11: Risk Profile Comparison

This figure compares predicted injury probabilities across low-risk, medium-risk, and high-risk profiles.

## Chapter 8: Discussion

### 8.1 Introduction

This chapter presents the most important results of the study, as well as interprets the results of the regression models and simulation analysis. The discussion relates the results of the research question and literature and discusses practical implications of research. The objective is to develop a clear picture on the effects of varying factors on the severity of road accidents and how such knowledge can be utilized in making improved road safety decisions.

### 8.2 Discussion of Environmental and Road Factors (RQ1)

The findings of the environmental and road context model (RQ1) reveal that the road conditions and infrastructure characteristics are significant factors in the severity of injuries. Variables like road surface condition and road alignment were both found to be important predictors in both model specifications.

Specifically, high risks of injury were related to the presence of sand, gravel, and muddy roads. Scenario-based simulation results further indicate that unsafe road surface conditions can increase the predicted probability of injury by approximately 36%, highlighting the critical importance of road maintenance and surface quality. Equally, complex and curved road alignments demonstrated higher injury risk unlike a straight road. The results imply that lack of vehicle control and stability in such conditions can be the causes of more dangerous consequences during the collisions.

Variables of traffic control also demonstrated a role in the probability of injuries. Situations in which there was no adequate control of the traffic or there were flashing lights were linked to increased risk of injuries. Simulation results suggest that the absence of proper traffic



control mechanisms can increase injury probability by approximately 17%, indicating that effective traffic management systems play a crucial role in reducing crash severity.

These observations are also aligning with past researches, which point to the fact that poor road infrastructure and unfavorable weather conditions contribute to the possibility of serious crashes. In general, the findings validate the fact that environmental and road-related factors play a significant role in the severity of injuries.

### 8.3 Discussion of Human and Vehicle Factors (RQ2)

The human and vehicle model findings reveal that human factors contribute more to injury severity than the environmental factors. The most important predictor was road user type which was found to be significant among all the variables.

Pedestrians, bicycle and motorcycle riders were highly vulnerable to injury, with odds of injury being very high when compared to those of vehicle occupants. Scenario-based simulation results show that motorcyclists can experience an increase in injury probability of up to 68% compared to baseline road users. The reasons are mainly that these users are not physically safeguarded and as such they are highly exposed in case of collisions. This result is very consistent with the existing literature where vulnerable road users are continuously reported as the most vulnerable population in traffic accidents.

There were also certain safety related factors like helmet use and other safety equipment, which were noted as a significant predictor of injury. Nonetheless, it does not mean that safety devices pose a risk of injury. Rather, it is a confounding effect because the use of safety equipment is more likely among high-risk road users like motorcyclists. Hence, this relationship is affected by the kind of road user and not necessarily the immediate impact of the safety device.

The influence of vehicle-related variables on injury severity was comparatively smaller than that of human factors. However, certain vehicle types, particularly buses and heavy vehicles, were associated with a lower probability of injury. This may be attributed not only to their robust structural design, which offers enhanced occupant protection, but also to behavioral and regulatory factors. Drivers of such vehicles are more likely to adhere to traffic regulations, operate within prescribed speed limits, and engage in safer driving practices. Furthermore, the presence of surveillance and monitoring systems in these vehicles may promote compliance and cautious driving, thereby contributing to reduced injury severity.

Additionally, factors such as person age (P\_AGE) and vehicle age (VEHICLE\_AGE) showed a positive association with injury severity; however, their impact was relatively moderate compared to stronger predictors such as road user type and environmental conditions.

In general, the results suggest that the most significant factors that determine the degree of injuries are human traits, particularly the type of road users.

### 8.4 Discussion of Combined Model RQ3

The combined model that combines environmental, road, human, and vehicle-related variables exhibited the highest predictive performance compared to the other models. This proves that the severity of crashes depends on the combination of different factors and not an isolated factor.

The findings indicate that the conditions of the road infrastructure and human attributes play a combined role in the results of injuries. Indicatively, the risk of injury is high when vulnerable road users are involved in accidents in unsafe road conditions. This shows the multi-effect of risk factors.

The better performance of the combination model supports the hypothesis that a combination of multiple domains offers a deeper explanation of the severity of the crashes. This observation aligns with the earlier studies, which advocate the need to consider various interacting variables in the study of transportation safety.

### 8.5 Discussion of Binary vs Ordinal Models

The comparison of binary and ordinal regression models indicates that binary logistic regression has a superior predictive performance with reference to this dataset. Binary models had better accuracy, balanced accuracy and ROC-AUC values than ordinal models.

The ordinal regression models were also limited by intense class imbalance especially the extremely small number of fatal injuries. Consequently, the models could not effectively forecast the outcome of fatal injuries, with the forecasts being shifted to no-injury and injury groups.

This shows that ordinal regression does not fit well when the number of rare categories of the outcome is not common. Conversely, binary classification is better and more consistent under these circumstances.

Thus, the findings indicate that binary logistic regression can be a more suitable model to investigate the severity of crash injury in this dataset.

### 8.6 Discussion of Simulation and Monte Carlo Analysis

The simulation analysis gives further information on the variation in injury probability across a variety of crash conditions. It was demonstrated with the help of scenario-based simulation that the number of injuries grows considerably with the introduction of high-risk conditions.

It refers to the results that high-risk road surface conditions, lack of traffic control, and presence of vulnerable road users result in high probability of injuries. Specifically, the risks

increased most when it came to situations in which motorcyclists were involved in accidents and the road was not safe.

Sensitivity analysis has also shown that other factors like age, vehicle age, and sex, although have some effect on the probability of injury, their roles are comparatively smaller than those of road user type and road conditions. As an example, Sex-related differences indicate that male drivers have a lower predicted probability of injury compared to female drivers; however, this effect is smaller relative to structural factors such as road user type and environmental conditions.

The Monte Carlo simulation demonstrated how the risk of injury may vary in various situations. Although majority of the simulated cases are moderate risk level, the percentage of very high injury probability is less. This proves that the crash risk is not homogenous and it varies according to an aggregation of various factors.

Generally, the findings presented in the simulation support the need to study individual and combined effects of variables when examining crash severity.

### 8.7 Key Findings

The results of this study provide few valuable lessons about road safety:

- Vulnerable road users such as pedestrians, bicyclists, and motorcyclists are at significantly higher risk of injury, with motorcyclists showing up to 68% higher injury probability in simulation analysis.
- Unsafe road conditions and poor infrastructure can increase injury probability by approximately 36%, emphasizing the importance of road maintenance.
- Lack of effective traffic control systems can increase injury probability by around 17%, highlighting the role of traffic management.

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- The severity of crashes depends on an interaction of a number of factors rather than one factor.
- Binary logistic regression is more efficient than ordinal regression where crash severity data is imbalanced.
- The simulation analysis demonstrates that the risk of injuries may grow exponentially when a combination of high-risk factors take place.

These lessons can assist policy makers and transportation planners to formulate specific safety interventions like road improvements, enhancing the effectiveness of traffic control systems, and offering more protection to the vulnerable road users.

## Chapter 9: Policy Recommendations

### 9.1 Introduction

This section contains a list of some practical and evidence-based recommendations based on the outcomes of the analysis and the results of the simulation conducted in the course of the research. The findings indicate the major risk factors that determine the severity of injuries in road accidents, such as environmental factors, road networks, and human attributes.

The recommendations will help policymakers and transportation authority's create specific interventions that would be effective in minimizing the severity of injuries. Importantly, the recommendations are based on quantitative evidence, which guarantees that the policy implications are consistent with the evidence.

### 9.2 Safety and Vulnerable Road Users

The results have shown that vulnerable users like pedestrians, bicycle users and motorcycle users are extremely exposed to injury risks compared to the occupants of the vehicle. The results of the simulation in the form of scenarios indicate that motorcyclists may be exposed to the risk of injury increase up to 68%, which demonstrates their high vulnerability.

Thus, the policymakers must focus on safety precautions that would specifically address the following groups:

- Creation of special lanes for cycles and motorcycles.
- Better walking paths, pavements and protective barriers.
- Enhanced educational activities on vulnerable road users.
- Increased implementation of safety laws like wearing of helmets and other protective equipment.

Urgent measures are needed to minimize the exposure and achieve better safety results regarding high-risk road users.

### 9.3 Improvement of Road Infrastructure

Unsafe road surface conditions and complicated road alignments were found out to be some of the key contributors to the severity of injuries. The findings of simulations demonstrated that poor road conditions may raise the risk of injury by 36%, which highlights the significance of the quality of infrastructures.

The authorities should pay attention to the following:

- Periodic repair of roads, particularly gravel and muddy roads and roads having surface damages.
- Better design of roads in the high-risk areas like sharp curves, slopes and intersections.
- Placing of warning signs and hazard indicators in risky areas.
- Constant checking of the roads by the use of modern technologies.

Enhancement of the quality of infrastructure can go a long way in minimizing the risk and impact of accidents in terms of injuries.

### 9.4 Targeted Safety Programs

The discussion indicates that various population groups are at risk of different levels. Although the human factor is the most dominant as far as causes of accidents are considered, this effect varies based on different categories like age, type of road user, and behavior.

Targeted safety programs should be developed for:

- Motorcyclists (safe riding and helmet awareness)
- Young drivers (training in responsible driving behavior)
- Senior road users (training and assistance with safe mobility)

- Risky geographic locations determined via accident data.

These targeted interventions may make safety interventions more effective, as they target vulnerable and high-risk groups.

### 9.5. Improving Traffic Control Systems.

It was found that the lack of traffic control or ineffectiveness of traffic control systems greatly contributed to the risk of injuries. The simulation findings show that injury probability can rise by about 17% when there is no adequate traffic control, which is why it is crucial to manage traffic properly.

In order to solve this problem, it is suggested to take the following measures:

- Traffic lights at high-risk intersections.
- Routine checking and repairing of the current traffic control systems.
- Installation of smart traffic control systems based on real-time information.
- Better road signs, road markings and visibility of lights.

Good traffic management mechanisms can be used to make the movement of vehicles manageable and minimize chances of major accidents.

### 9.6 Data-Driven Policy Making

This study has revealed that predictive modeling and simulation techniques are useful in the management and prediction of road safety risks. These tools help the policymakers to recognize the high-risk situations and can also assess the effectiveness of various interventions before they are implemented.

Government agencies need to use data-driven methods to:

- Determine the accident-prone areas and the risky environments.
- maximize use of resources to enhance road safety.
- Formulate evidence-based interventions and policies.



- Trial safety strategies using simulation models.

By leveraging analytics and simulation, policymakers can move from reactive to proactive decision-making in road safety management.

### 9.7 Summary of Policy Implications

Overall, the findings of this study emphasize that injury severity in road accidents is influenced by a combination of environmental, infrastructural, and human factors. It is evident that the results indicate that:

- Unsafe road conditions can increase injury probability by 36%
- Absence of traffic control systems can raise the chances of injuries by 17%
- Vulnerable road users, particularly motorcyclists, face up to 68% higher injury risk

These insights highlight the need for comprehensive and integrated road safety strategies that address multiple dimensions simultaneously.

In Conclusion, the recommendations contained in this chapter are closely related to the empirical results of the study and they provide a realistic way of ensuring the extent of injuries is reduced in road incidents. These measures can help to establish safer transportation systems and better public safety results.

## Chapter 10: Limitations and Future Work

Although this study offers useful information on the severity of crash injuries, it has some limitations that must be taken into consideration.

### 10.1 Limitation

#### 1) Data Limitations

The data used in the present study refers to the time frame between 1999 and 2014. Any modifications in road safety rules, technology and infrastructure that occur after this time are not reflected. Thus, the findings might not be entirely representative of the existing roads.

#### 2) Lost and Missing Variables.

Certain variables were summarized under the Other category because of missing or unavailable values. Though this is good in preserving data, some of the results will become less interpretable.

#### 3) Lack of Behavioral Variables

Other critical elements like driver distraction, alcohol use, speed and real time traffic conditions were not represented in the data. These factors might play a very important role in the severity of the crash.

#### 4) Class Imbalance

There are very little fatal injury cases in the dataset that influenced the performance of the ordinal regression model. This constraint limited the modelling of severity at all levels.

#### 1) Observational Data

The study is found on observational data, and this implies that the findings are relationships and not causal.

## 10.2 Future Work

This study can be improved in the future research by:

- Use more recent datasets to reflect current road conditions
- Add other variables including speed, distraction and geographic location.
- Use sophisticated machine learning algorithms like Gradient Boosting or Random Forest.
- Manage the imbalance in the classes with such techniques as resampling or weighted models.
- Give simulation analysis to real-time decision support systems.

Such enhancements have the potential to offer better understanding and improve the practice of crash severity analysis.

## Conclusion

This study has explored the factors that determine the magnitude of injury during road accidents in Canada through statistical modelling and simulation methodology. The National Collision Database was used to conduct the analysis and has numerous observations representing environmental, road, human, and vehicle-related factors.

Binary logistic regression model results indicate that environmental conditions and human characteristics influence the probability of being injured significantly. Road user type turned out to be the strongest of all the variables, with vulnerable road users, including pedestrians, bicyclists, and motorcyclists, having a much greater risk of injury.

The combined model was the most predictive, which proved that the severity of crashes is conditioned by the interplay of several factors and not only one variable. Conversely, ordinal regression models were limited because of the imbalance of classes especially in the prediction of fatal injuries.

Simulation and Monte Carlo analysis provided additional insights into how injury probability changes under different conditions. The findings indicate that the risk of injury is highest when a combination of high-risk factors is present particularly when the risk of road conditions and vulnerable road users is involved.

Overall, this study highlights the importance of integrated approaches to road safety, combining infrastructure improvements, human behavior awareness, and data-driven policy making. These results can be used to assist transportation planners and policymakers to develop effective measures to minimize the severity of injuries in road accidents.

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This study contributes to the field of transportation safety by combining statistical modeling with simulation techniques, providing both predictive insights and practical applications for improving road safety.

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## Appendices

### Appendix A: Raw Data

The dataset used in this study is the **National Collision Database (NCDB)** provided by Transport Canada, covering road accident records from 1999 to 2014.

Due to accessibility limitations of the official source at the time of submission, the dataset was accessed through a secondary platform (Kaggle) for research purposes.

The dataset contains over 5.8 million records and includes variables related to:

- Environmental conditions
- Road characteristics
- Traffic control
- Human factors
- Vehicle attributes

### Appendix B: Supporting Documentation

This appendix provides supporting materials related to the methodology, modeling process, and analysis conducted in this study.

#### B1: Python Code Files

The analysis was conducted using Python. The following components were implemented:

- Data preprocessing and cleaning scripts
- Binary logistic regression models
- Ordinal regression models
- Scenario simulation scripts
- Monte Carlo simulation code



## COLLISION SEVERITY ANALYSIS IN CANADIAN ROAD ACCIDENTS

### B2: Data Visualization

All visualizations presented in this report were generated using Python libraries and analytical tools. These include:

- Model performance comparison charts
- Odds ratio visualizations (risk and protective factors)
- Scenario and sensitivity analysis plots
- Monte Carlo simulation distributions and risk bands

### B3: Software and Libraries Used

- Python (Pandas, NumPy, Statsmodels, Scikit-learn, Matplotlib)
- Microsoft Excel

### B4: Additional Outputs

Additional intermediate results, model outputs, and validation steps were generated during the analysis but are not included in the main report for brevity.